LAB 7: Configuration of Switch, VLAN Configuration and Inter-VLAN Routing

OBJECTIVES:

1. To understand the concept of Virtual Local Area Networks (VLANs) and their importance in network segmentation.
2. To configure VLANs on a Cisco switch and assign specific ports to different VLANs.
3. To configure trunking using IEEE 802.1Q protocol.
4. To implement Inter-VLAN routing using the Router-on-a-Stick method.
5. To verify communication between devices in different VLANs.

THEORY:

1. VLAN (Virtual Local Area Network)

A VLAN is a logical grouping of devices within the same broadcast domain, regardless of their physical location on a switch. VLANs allow network segmentation at Layer 2.

Operation:
A physical switch is divided into multiple logical networks. Devices assigned to one VLAN cannot communicate directly with devices in another VLAN without a Layer 3 device such as a router.

Advantages:

- Improves security
- Reduces broadcast traffic
- Organizes users by department rather than physical location
- Enhances network performance

2. Trunking (IEEE 802.1Q)

Trunking allows multiple VLANs to share a single physical link between network devices (e.g., switch to router).

Operation:
It uses the IEEE 802.1Q tagging standard, which inserts a VLAN tag into Ethernet frames. This tag identifies which VLAN the frame belongs to.

Trunk links are required when multiple VLANs need to communicate through a router.

3. Inter-VLAN Routing (Router-on-a-Stick)

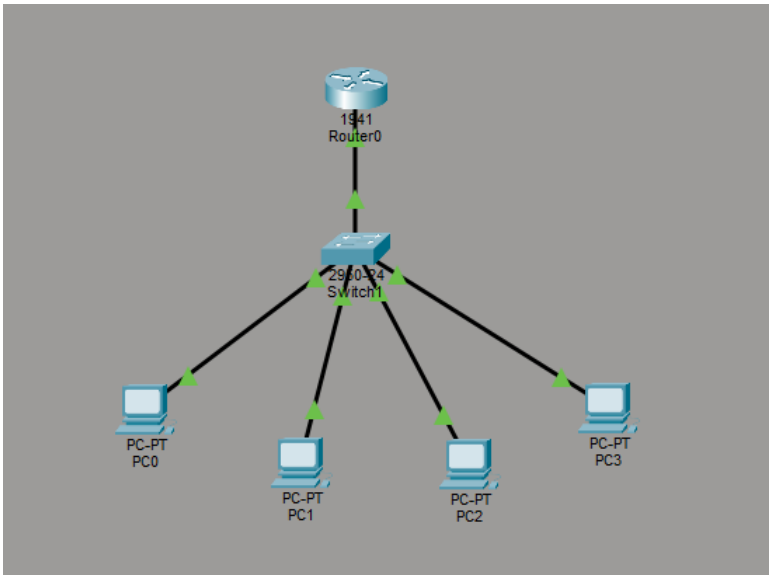
Since VLANs are separate broadcast domains, devices in different VLANs cannot communicate without routing.

Operation:
A single physical interface on the router is divided into multiple sub-interfaces.
Each sub-interface:

- Is assigned to a specific VLAN
- Uses 802.1Q encapsulation
- Has an IP address that acts as the Default Gateway for that VLAN

This method is called **Router-on-a-Stick**.

NETWORK TOPOLOGY:



- One **Router0 (1941)** is connected to a **Cisco 2950-24 Switch**.
- The switch connects four PCs.

VLAN Assignment:

- **VLAN 10 (Name: computer)**
  - PC0
  - PC1
- **VLAN 20 (Name: Electronics)**
  - PC2
  - PC3

The connection between the switch and router is configured as a **Trunk link** to carry traffic for both VLANs.

IP CONFIGURATION TABLE:

Device	Interface	IP Address	Subnet Mask	Default Gateway	VLAN
Router0	G0/0.10	192.168.10.1	255.255.255.0	N/A	10
Router0	G0/0.20	192.168.20.1	255.255.255.0	N/A	20
PC0	FastEthernet0	192.168.10.2	255.255.255.0	192.168.10.1	10
PC1	FastEthernet0	192.168.10.3	255.255.255.0	192.168.10.1	10
PC2	FastEthernet0	192.168.20.2	255.255.255.0	192.168.20.1	20
PC3	FastEthernet0	192.168.20.3	255.255.255.0	192.168.20.1	20

CONFIGURATION:

1. Switch Configuration Commands (VLANs and Ports)

```
Switch(config)# hostname Arpan
Arpan(config)# vlan 10
Arpan(config-vlan)# name computer
Arpan(config-vlan)# exit
Arpan(config)# vlan 20
Arpan(config-vlan)# name Electronics
Arpan(config-vlan)# exit
```

Assigning Ports to VLANs:

```
Arpan(config)# interface range fa0/2-3
Arpan(config-if-range)# switchport mode access
Arpan(config-if-range)# switchport access vlan 10
Arpan(config-if-range)# exit

Arpan(config)# interface range fa0/4-5
Arpan(config-if-range)# switchport mode access
Arpan(config-if-range)# switchport access vlan 20
Arpan(config-if-range)# exit
```

Configuring Trunk Link to Router:

```
Arpan(config)# interface fa0/1
Arpan(config-if)# switchport mode trunk
```

2. Router Configuration Commands (Inter-VLAN Routing)

Configuring Sub-interface for VLAN 10:

```
Router0(config)# interface g0/0.10
Router0(config-subif)# encapsulation dot1q 10
Router0(config-subif)# ip address 192.168.10.1 255.255.255.0
```

Configuring Sub-interface for VLAN 20:

```
Router0(config)# interface g0/0.20
Router0(config-subif)# encapsulation dot1q 20
Router0(config-subif)# ip address 192.168.20.1 255.255.255.0
```

TESTING AND VERIFICATION

After configuration:

- The command `show vlan brief` was used to verify VLAN creation and port assignments.
- VLAN 10 displayed "computer".
- VLAN 20 displayed "Electronics".

```
Switch#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/6, Fa0/7, Fa0/8, Fa0/9
                                                Fa0/10, Fa0/11, Fa0/12, Fa0/13
                                                Fa0/14, Fa0/15, Fa0/16, Fa0/17
                                                Fa0/18, Fa0/19, Fa0/20, Fa0/21
                                                Fa0/22, Fa0/23, Fa0/24
10   computer                         active    Fa0/2, Fa0/3
20   Electronics                      active    Fa0/4, Fa0/5
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
Switch#
```

Ping testing was performed:

- Ping from **PC0 (192.168.10.2)** to **PC1 (192.168.10.3)** → Successful (Same VLAN).
- Ping from **PC0 (VLAN 10)** to **PC2 (VLAN 20)** → Successful after Inter-VLAN routing configuration.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.3

Pinging 192.168.10.3 with 32 bytes of data:

Reply from 192.168.10.3: bytes=32 time=7ms TTL=128
Reply from 192.168.10.3: bytes=32 time=3ms TTL=128
Reply from 192.168.10.3: bytes=32 time<1ms TTL=128
Reply from 192.168.10.3: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 7ms, Average = 2ms

C:\>
```

RESULT:

The VLANs were successfully created on the switch, and ports were correctly assigned.
Trunking was properly configured using IEEE 802.1Q.
Inter-VLAN routing was successfully implemented using the Router-on-a-Stick method.

Devices within the same VLAN communicated directly, while devices in different VLANs communicated through the router.

DISCUSSION AND CONCLUSION:

During this lab session, VLANs were implemented to logically segment the network into two separate departments: "computer" and "Electronics." Initially, devices in different VLANs could not communicate because VLANs form separate broadcast domains.

By configuring trunking and implementing Inter-VLAN routing using Router-on-a-Stick with 802.1Q encapsulation, communication between VLAN 10 and VLAN 20 was successfully established.

This experiment provided practical understanding of:

- Network segmentation
- Broadcast domain isolation
- VLAN tagging
- Router sub-interfaces
- Inter-VLAN communication

Hence, the lab was completed successfully with proper knowledge and implementation of VLAN configuration and Inter-VLAN routing using Cisco Packet Tracer.